13/08/24

Algorithm (PB)

① Long integer multiplication

② Finding the closest pair of points

→ ① usual time complexity, $O(n^2)$ for n digits

A & B are 2 n digits numbers

$A = [n-1, \ldots, 0]$ $\quad$ $B = [n-1, \ldots, 0]$

	$\leftarrow n/2 \rightarrow$	$\leftarrow \frac{n}{2} \rightarrow$
A =	w	x
B =	y	z
	wz	xz
wy	xy	

divide until single element

$m = n/2 \quad A = w \cdot 10^m + x \quad B = y \cdot 10^m + z$

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$A \cdot B = (w \cdot 10^m + x)(y \cdot 10^m + z)$

$= w \cdot y \cdot 10^m + (w \cdot z + x \cdot y) \cdot 10^m + xy$

$\Rightarrow$ 4 multiplications

shift m times $\Rightarrow 10^m = \Theta(n)$

3 additions $\Rightarrow \Theta(n)$

Algorithm

mult(A, B):

if $m == 1$

find out the product

else {

$$c = \begin{cases} \text{mult}(w, y, n/2) \\ \text{mult}(w, z, n/2) \\ \text{mult}(x, y, n/2) \\ \text{mult}(x, z, n/2) \end{cases}$$

}

return c

$T(n) = 4T(n/2) + \Theta(n)$

$a = 4, b = 2, k = 1$

$= \Theta(n^{\log_2 4}) = \Theta(n^2)$

no gain so far

faster Algo implementation with algebric trick

$C = w \cdot y \qquad D = x \cdot z \qquad E = (w+x)(y+z) - C - D$

$= wy + xy + wz + xz$

$A \cdot B = wy\,10^{2m} + (wz + xy)\,10^m + xy$

$E - C - D = (wy + xy + wz - xz) - wy - xy$

$\therefore A \cdot B = C \cdot 10^{2m} + E \cdot 10^m + D$

$T(n) = 3T(n/2) + \Theta(n)$

$= \Theta(n^{\log_2 3})$

$= \Theta(n^{1.585})$

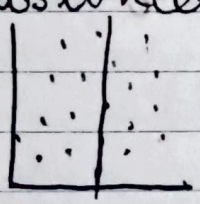
2) Closest Pair of points calculation

given a set of points

Given a set P of n points (x, y), find the pair of points having minimum Euclidean distance $(x_1, y_1), (x_2, y_2) \Rightarrow \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

Naive Algo: find distance between n_{c_2} ~~points~~ pairs & find the minimum distance. Time complexity $O(n^2)$

Aim: $n \log n$



Basis: if $|P| \leq 3$, solve by brute force in constant time.

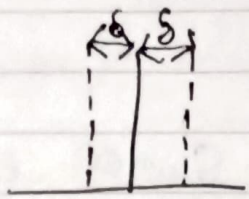
Divide:

Partition the points into subarrays P_L and P_R based on their x coordinate (Assuming points are already sorted on x).

Conquer:

compute closest pair within P_L and P_R Neceeryly. Let δ_L and δ_R be the corresponding minimum distance.

Let $\delta = \min(\delta_L, \delta_R)$



<u>Combine</u>:

The closest point $p \in P_R$ & $q \in P_q$ need to be obtained in $O(n)$ time

Let S_y: set of points in 2δ region across l sorted on y coordinates (presorted).

<u>Lemma</u>: Given any 2 points $s_i, s_j \in S_y$, if $dist(s_i, s_j) \leq \delta$, then $|j-i| \leq 7$

- Define a square of length $\delta/2$ in the 2δ region resulting in such seperate

- maximum distance between the pair of points in a square = len of diagonal

$= \sqrt{(\delta/2)^2 + (\delta/2)^2} = \delta/\sqrt{2}$

$\delta/\sqrt{2} < \delta$

Since each square lies entirely on one side of L, no square can contain two or more points of P (if they were there, they would have caught during recursion)

→ there can be atmost eight points of S in the rectange one for each square, that are candidate for closest pairs with less than δ distances. & thus $|j-i| \leq 7$

1. Sort x $n \log n$

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2. Sort y $n \log n$

10 3. D&C $\longrightarrow$ $T(n) = 2T(n/2) + \Theta(n)$

$= \Theta(n \log n)$

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